

Grade 11 Functions

Hyperbolic Function

A hyperbolic function is NOT polynomial. It is a function representing a constant ratio. For example you could consider the function $xy = 4$. This is a function of all the $(x; y)$ pairs that multiply together to give the value 4. We usually write this as $y = \frac{4}{x}$.

Last year we studied hyperbolic functions of the form: $y = \frac{a}{x} + q$. These would be the $(x; y)$ pairs that multiply to give a if you adjust the y co-ordinate by q units.

This year we will be studying hyperbolic functions of the form: $y = \frac{a}{x+p} + q$.

Use Geogebra to open the “Grade 11 Hyperbola Investigation”.

Web Address: <http://tube.geogebra.org/student/m968781>

Offline web-page: [S:\Grade11\Maths and APM\Functions Investigation](#) – “Investigation 2 – Hyperbolic Function”

Geogebra file: [S:\Grade11\Maths and APM\Functions Investigation\Geogebra Files](#) – “Grade 11 Hyperbola Investigation – Geogebra file”

Once you have Geogebra running you should see a Cartesian plane, a blue line running across the screen and 3 sliders, labelled a , p and q and a checkbox labelled “Show axes of symmetry”. You should also see the equation of the function written in blue. **Ensure that a , p and q are all set to 0 and that the checkbox is NOT ticked.**

1. When dealing with hyperbolas it is important to understand what an asymptote is. Define an asymptote.

2. Keep $a = 0$. The equation should be $f(x) = \frac{0}{x}$. Is the function a straight line that is identical to the one we saw when looking at parabolas (for $f(x) = 0x^2$)? Explain your answer

3. Keep $a = 0$. Does this function have a horizontal asymptote?

4. Adjust the value of a so that $a > 0$.

a) Describe what happens to the graph.

b) What is the difference in the graph when a is close to 0 and when a is close to 5?

5. Repeat step 4, but this time adjust the value of a so that $a < 0$.

a) Describe what happens to the graph.

b) What is the difference in the graph when a is close to 0 and when a is close to -5 ?

6. Ensure that a is NOT 0. Keep $p = 0$. Adjust the value of q .

a) Describe how the function text changes (the text is in blue underneath the sliders).

b) Describe how the graph (the picture) changes as q changes. Be precise as to how the graph changes when you increase q and when you decrease q .

7. Ensure that a is NOT 0. Adjust q so that $q = 0$. Adjust the value of p so that $p > 0$.

a) Describe how the function text change as you increase p . (Make p bigger than 0).

b) Describe how the graph changes as you increase p .

c) Is the change as you increase p similar to the change you saw in a parabola? Describe the similarity / difference?

8. Ensure that a is NOT 0. Adjust the value of p so that $p < 0$.

a) Describe how the function text change as you decrease p . (Make p less than 0).

b) Describe how the graph changes as you decrease p .

c) Now adjust a so that $a < 0$. Change the slider for p . Does the sign of a make a difference to the way that p changes the function?

9. Pick any values of a, p and q so long as a is not 0.

a) Look at the horizontal and vertical asymptotes.

b) Look at the values of p and q .

c) How do the asymptotes link to the values of p and q ?

10) Pick other values of a, p and q . Repeat 9a to 9c.

11) Write down 2 properties of a quadratic function that a changes. Describe how they change as a changes.

12) Write down 2 properties of a quadratic function that p changes. Describe how they change as p changes.

13) Write down 2 properties of a quadratic function that q changes. Describe how they change as q changes.

14. Pick any values of a , p and q so long as a is not 0. Tick the box called show axes of symmetry. You should see two green lines appear

a) What is an axis of symmetry?

b) Look at the values of p and q . Look at the point of intersection of the two axes.

c) How does the intersection of the two axes link to the values of p and q ?

d) Look at the gradient of the axes. What are the two gradients. Adjust the values of a , p and q . Does the gradient change?

e) Determine the equations of the axes of symmetry.

Homework – sketching a parabola of the form $y = a(x + p)^2 + q$

1. Pick any 3 values of a , p and q .
2. Without using Geogebra or a calculator plot the turning point of the function
3. Without using Geogebra or a calculator plot the y -intercept of the function
4. Without using Geogebra plot the x -intercepts of the function. You can use a calculator to help you solve the equation if necessary.
5. Draw the function, including the asymptotes.
6. Determine the equations of the axes of symmetry.
7. Check your answer using Geogebra. You can check the equations of the axes of symmetry by typing the equations you came up with into Geogebra and seeing if they are drawn correctly.

Repeat steps 1-6 above so that you have drawn 4 of your own functions. Use different combinations of positive and negative a , p and q values.